Title:

TrueSay

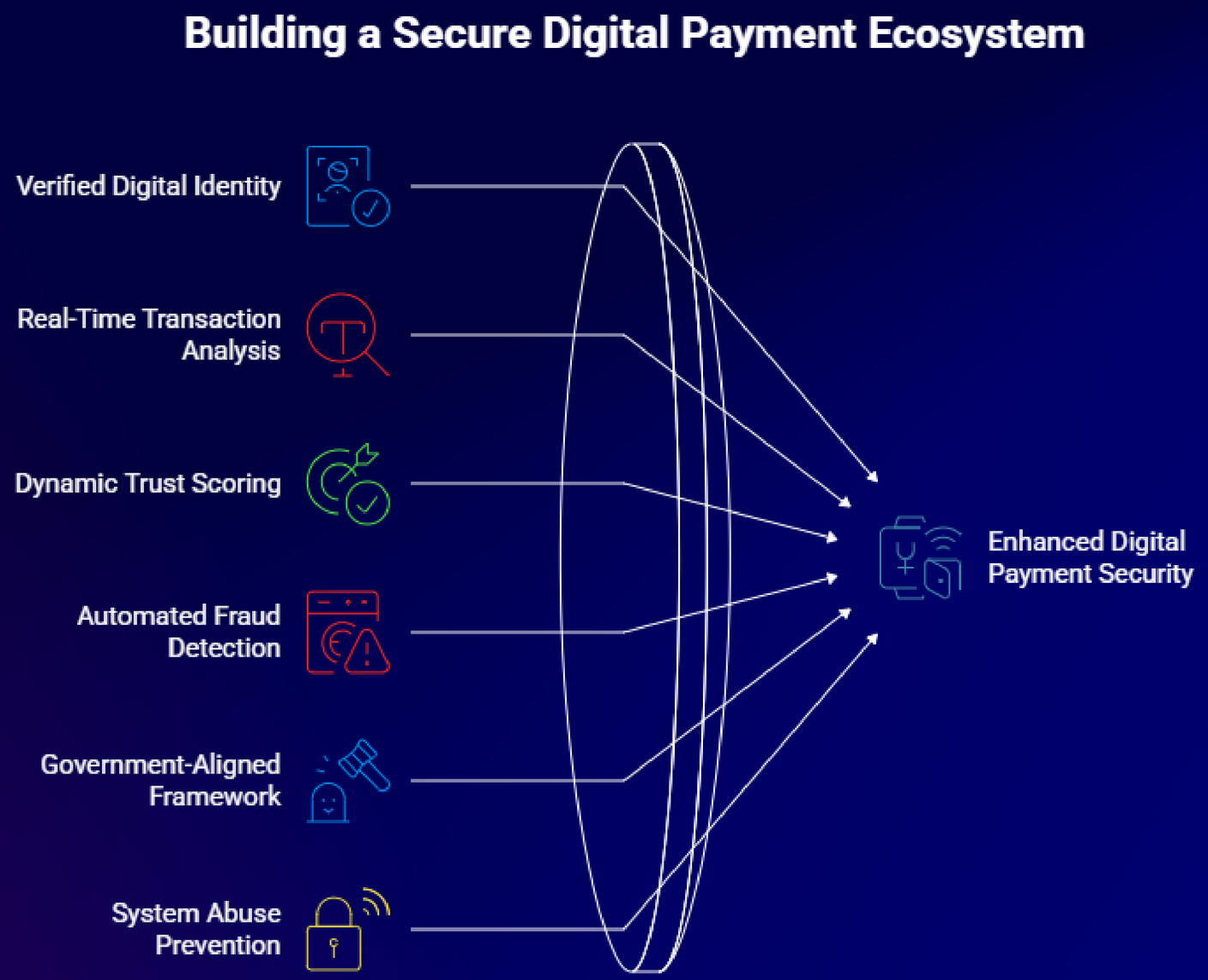
security is our key priority

Aim:

Develop an AI-powered real-time fraud detection system that analyzes digital transactions, assigns a dynamic Trust Score to users, and instantly detects and prevents fraudulent activity to ensure secure and safe payments.

Objective:

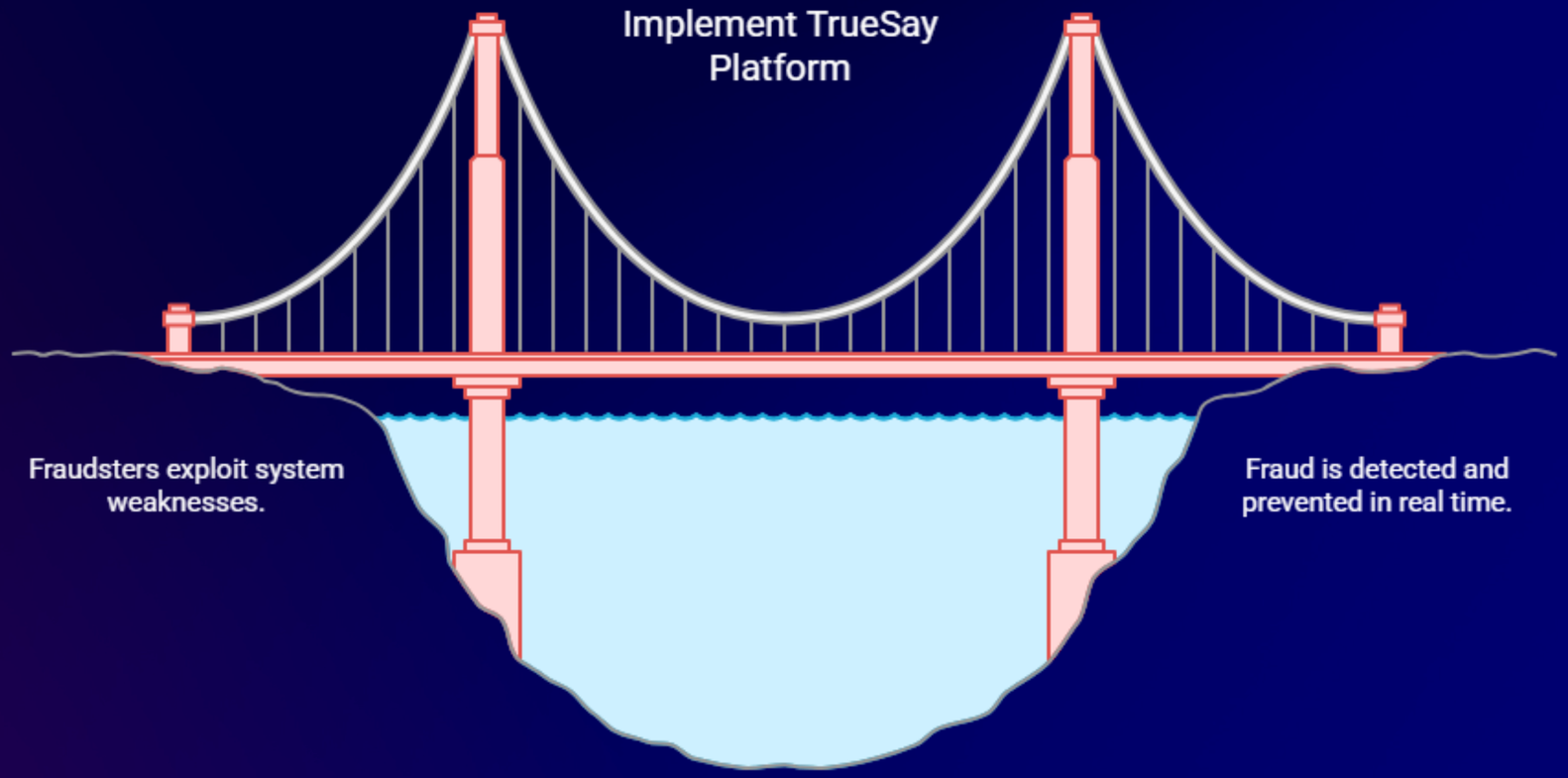
- Verified Identity: Users must register with a verified digital identity before making transactions. Unverified accounts have limited payment access.
- Real-Time Monitoring: Every transaction is analyzed in real time based on behavior, location, and transaction history to detect suspicious activity.
- Dynamic Trust Score: Each account receives a Trust Score (0–100) that updates after every transaction; low scores trigger restrictions or delays on large payments.
- Automatic Fraud Alerts: Suspicious transactions instantly notify the user and bank, allowing the user to confirm or block the payment.
- Regulatory Action Framework: Fraud actions follow government and Reserve Bank of India (RBI) regulations with a structured penalty process.
- Abuse Prevention: Accounts have two “lives”; repeated confirmed fraud leads to permanent termination and blocked re-entry.
- Emergency Support: The system provides quick access to fraud helplines and an AI chatbot to guide users in reporting and staying safe.



Problem Statement:

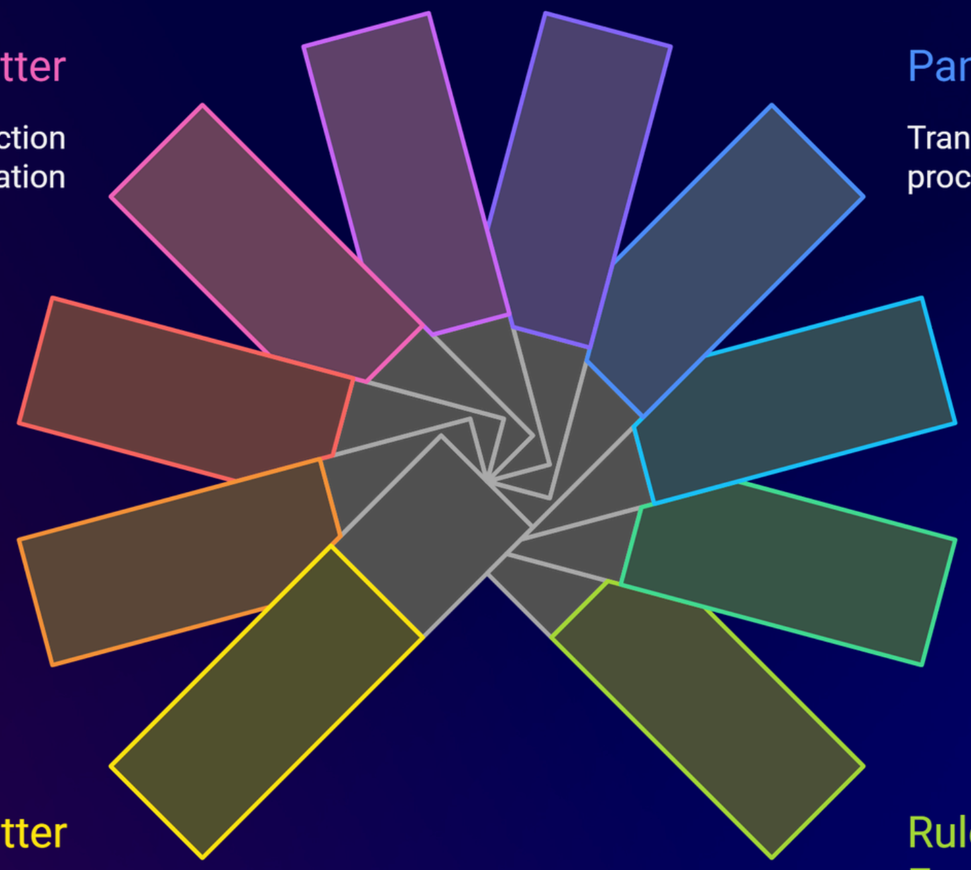
- Digital Payment Growth: Systems like Unified Payments Interface (UPI) and online banking have made transactions fast and widely accessible.
- Increasing Fraud: Cyber threats such as phishing, identity theft, and unauthorized transactions are rising.
- Weak Traditional Detection: Rule-based systems cannot effectively detect advanced fraud.
- Need for Solution: A real-time AI platform that assigns a Trust Score, analyzes transactions, and prevents fraud while following **Reserve Bank of India regulations.

TrueSay bridges the gap between digital payments and fraud prevention.



TechStack:

Tech stack layers from user interface to core logic.



Python FastAPI

High-performance REST API server

Scikit-learn

Machine learning for risk scoring

Charts Flutter

Live transaction visualization

Pandas / NumPy

Transaction data processing

Firebase SDK

Real-time data binding

JWT Authentication

Secure token-based session management

Dart

Application logic and UI interaction

GPT-4o API

Generates plain-English explanations

Flutter

Cross-platform mobile application

Rule-Based Scoring Engine

Real-time scoring across signals

User Interface

Core Logic